



Exploratory Data Analysis On E-Commerce Sales

#Goal:-Analyze sales trends, customer behavior, category performance, and city-wise revenue to generate business insights.

```
In [1]: import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
plt.style.use('ggplot')
sns.set_palette('deep')
```

```
In [2]: df=pd.read_csv(r"C:\Users\Sahil Singh\Sam_Analysis\03_Project_E-Commerce_D
```

Understanding Dataset

```
In [3]: #Displays first 10 rows.
df.head(10) # Dataset contains order-level sales information.
```

Out[3]:	Order ID	Order Date	Customer Name	City	Category	Product Name	Quantity	Unit Price	
0	10001	10/19/2024	Kashvi Varty	Bangalore	Books	Non-Fiction Ipsum	2	36294	C
1	10002	8/30/2025	Advik Desai	Delhi	Groceries	Rice Nemo	1	42165	S
2	10003	11/4/2023	Rhea Kalla	Patna	Kitchen	Juicer Odio	4	64876	D
3	10004	5/23/2025	Anika Sen	Kolkata	Groceries	Oil Doloribus	5	37320	C
4	10005	1/19/2025	Akarsh Kaul	Pune	Clothing	Kids Wear Quo	1	50037	
5	10006	4/6/2025	Vardaniya Jayaraman	Pune	Furniture	Chair Assumenda	5	40287	
6	10007	10/13/2023	Drishya Khare	Mumbai	Clothing	Accessories Minima	2	3636	
7	10008	8/17/2025	Misha Dua	Lucknow	Books	Biography Vel	1	15885	S
8	10009	3/7/2025	Arhaan Vala	Jaipur	Groceries	Spices Expedita	1	40834	C
9	10010	12/2/2024	Lavanya Hayer	Goa	Kitchen	Juicer Voluptatibus	2	16686	

In [4]: `#Check last rows`
`df.tail()`

Out[4]:

	Order ID	Order Date	Customer Name	City	Category	Product Name	Quantity	Unit Price
4995	14996	6/25/2024	Nishith Kulkarni	Kolkata	Books	Fiction Veritatis	3	6067
4996	14997	12/22/2024	Aaina Chander	Jaipur	Toys	Doll Nulla	5	7004
4997	14998	4/15/2025	Dhanush Gara	Bangalore	Beauty	Lipstick Eaque	1	4216
4998	14999	7/8/2024	Divyansh Malhotra	Kolkata	Electronics	Smartwatch Adipisci	4	1356
4999	15000	2/4/2024	Aarush Walla	Goa	Clothing	Kids Wear Repellat	1	7676

In [5]: `#Check Statistical Summery`
`df.describe(include='all')` `#Total price distribution appears right-skewed`

Out[5]:

	Order ID	Order Date	Customer Name	City	Category	Product Name	Quantity
count	5000.000000	5000	5000	5000	5000	5000	5000.000000
unique	NaN	730	4844	20	10	3835	NaN
top	NaN	12/31/2023	Vritika Cherian	Guwahati	Books	Chair Adipisci	NaN
freq	NaN	15	3	293	528	5	NaN
mean	12500.500000	NaN	NaN	NaN	NaN	NaN	2.992600
std	1443.520003	NaN	NaN	NaN	NaN	NaN	1.413100
min	10001.000000	NaN	NaN	NaN	NaN	NaN	1.000000
25%	11250.750000	NaN	NaN	NaN	NaN	NaN	2.000000
50%	12500.500000	NaN	NaN	NaN	NaN	NaN	3.000000
75%	13750.250000	NaN	NaN	NaN	NaN	NaN	4.000000
max	15000.000000	NaN	NaN	NaN	NaN	NaN	5.000000

In [6]: `#Check data_types,Null_values,Memory_usage`
`df.info()`

```

<class 'pandas.DataFrame'>
RangeIndex: 5000 entries, 0 to 4999
Data columns (total 10 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Order ID              5000 non-null   int64
1   Order Date            5000 non-null   str
2   Customer Name         5000 non-null   str
3   City                  5000 non-null   str
4   Category              5000 non-null   str
5   Product Name          5000 non-null   str
6   Quantity              5000 non-null   int64
7   Unit Price            5000 non-null   int64
8   status                5000 non-null   str
9   Payment Mode          5000 non-null   str
dtypes: int64(3), str(7)
memory usage: 740.4 KB

```

```

In [7]: #Checks column names.
        df.columns

```

```

Out[7]: Index(['Order ID', 'Order Date', 'Customer Name', 'City', 'Category',
              'Product Name', 'Quantity', 'Unit Price', 'status', 'Payment Mode'],
              dtype='str')

```

Data Cleaning..

```

In [8]: #Missing Values
        df.isnull().sum() #None In this DataSet

```

```

Out[8]: Order ID          0
        Order Date       0
        Customer Name    0
        City             0
        Category         0
        Product Name     0
        Quantity         0
        Unit Price       0
        status           0
        Payment Mode     0
        dtype: int64

```

```

In [9]: # Duplicate Values
        df.duplicated().sum() # None in this dataset

```

```

Out[9]: np.int64(0)

```

```

In [10]: # Dropping columns
         df.drop('Order ID',axis=1,inplace=True)

```

```

In [11]: # Standardize the data

```

```
df.columns=df.columns.str.lower().str.strip().str.replace(' ','_')
df=df.rename(columns={'unit_price':'price'})
df.columns
```

```
Out[11]: Index(['order_date', 'customer_name', 'city', 'category', 'product_name',
               'quantity', 'price', 'status', 'payment_mode'],
              dtype='str')
```

```
In [12]: # Fixing Data types
# Here, Order date columns show Str
df['order_date']=pd.to_datetime(df['order_date'], format="mixed")
df['order_date'].dtype
```

```
Out[12]: dtype('<M8[us]')
```

```
In [13]: # Feature Engineering
df['order_month']=df['order_date'].dt.month_name()
df['order_year']=df['order_date'].dt.year
df['order_month'].head()
df['order_year'].head()
```

```
Out[13]: 0    2024
1    2025
2    2023
3    2025
4    2025
Name: order_year, dtype: int32
```

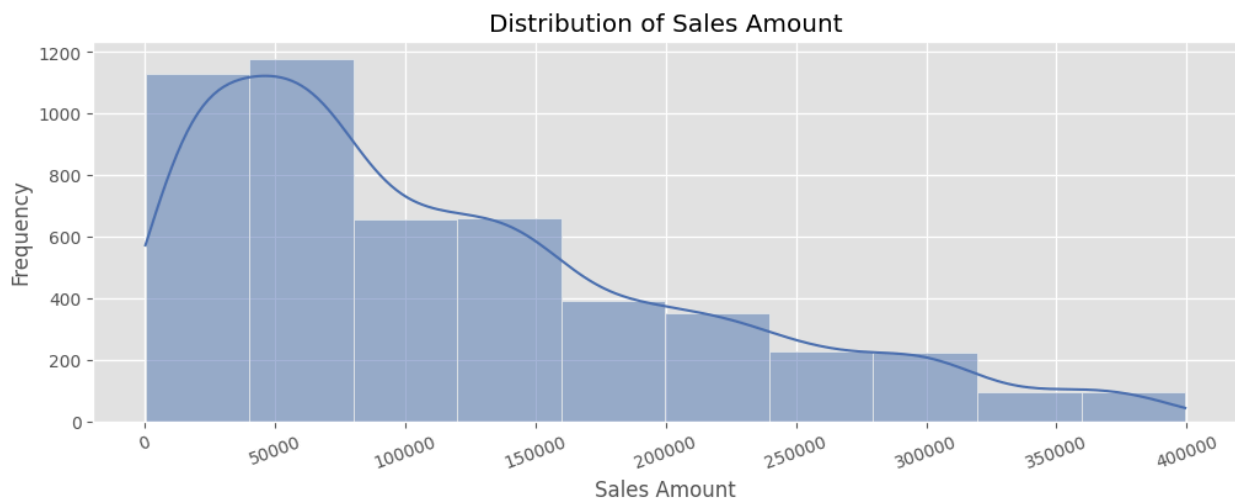
```
In [14]: df['total_price']=df['price']*df['quantity'] #Important for KPIs
df['total_price'].head()
```

```
Out[14]: 0    72588
1    42165
2   259504
3   186600
4    50037
Name: total_price, dtype: int64
```

Analysis

```
In [15]: #Univariate Analysis
```

```
In [16]: plt.figure(figsize=(12,4))
sns.histplot(df['total_price'],bins=10,kde=True)
plt.title('Distribution of Sales Amount')
plt.xlabel('Sales Amount')
plt.ylabel('Frequency')
plt.xticks(rotation=20)
plt.show()
```

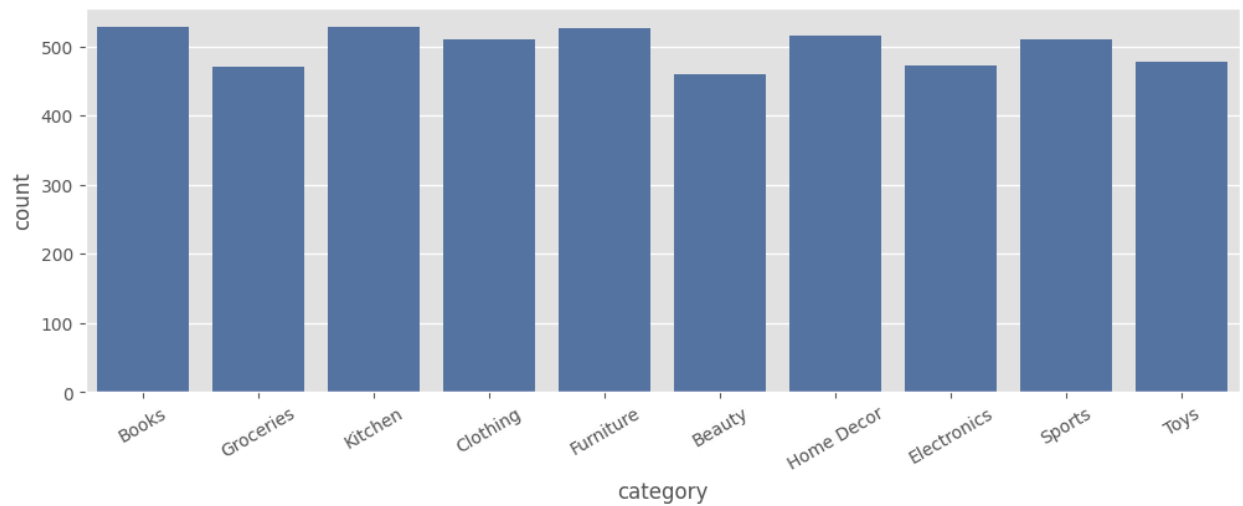


```
In [17]: #Insight  
#Most orders are low-value, Few orders are high-value & Right-skewed distr
```

```
In [18]: #Check Category Frequency  
df['category'].value_counts()
```

```
Out[18]: category  
Books          528  
Kitchen        528  
Furniture      527  
Home Decor     515  
Clothing       511  
Sports         511  
Toys           478  
Electronics    472  
Groceries      470  
Beauty         460  
Name: count, dtype: int64
```

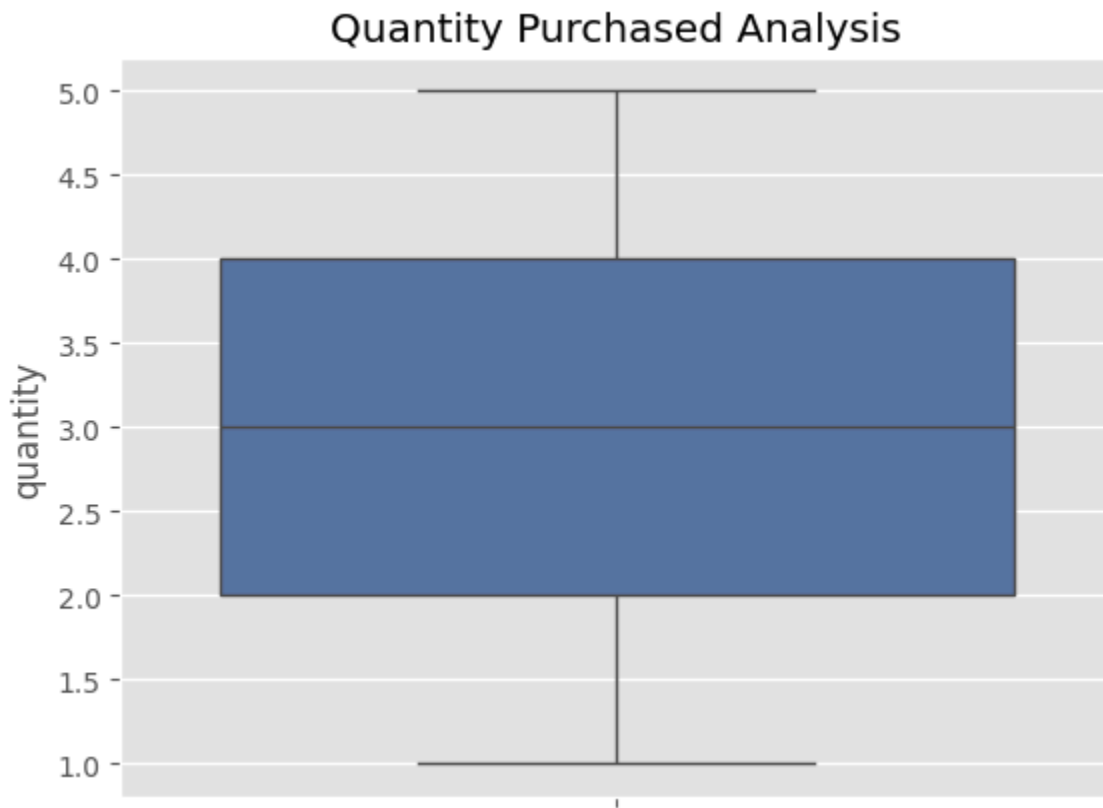
```
In [19]: plt.figure(figsize=(12,4))  
sns.countplot(x='category', data=df)  
plt.xticks(rotation=30)  
plt.show()
```



```
In [20]: #Insight  
#Shows which product categories dominate sales volume.
```

```
In [21]: sns.boxplot(df['quantity'])  
plt.title('Quantity Purchased Analysis')
```

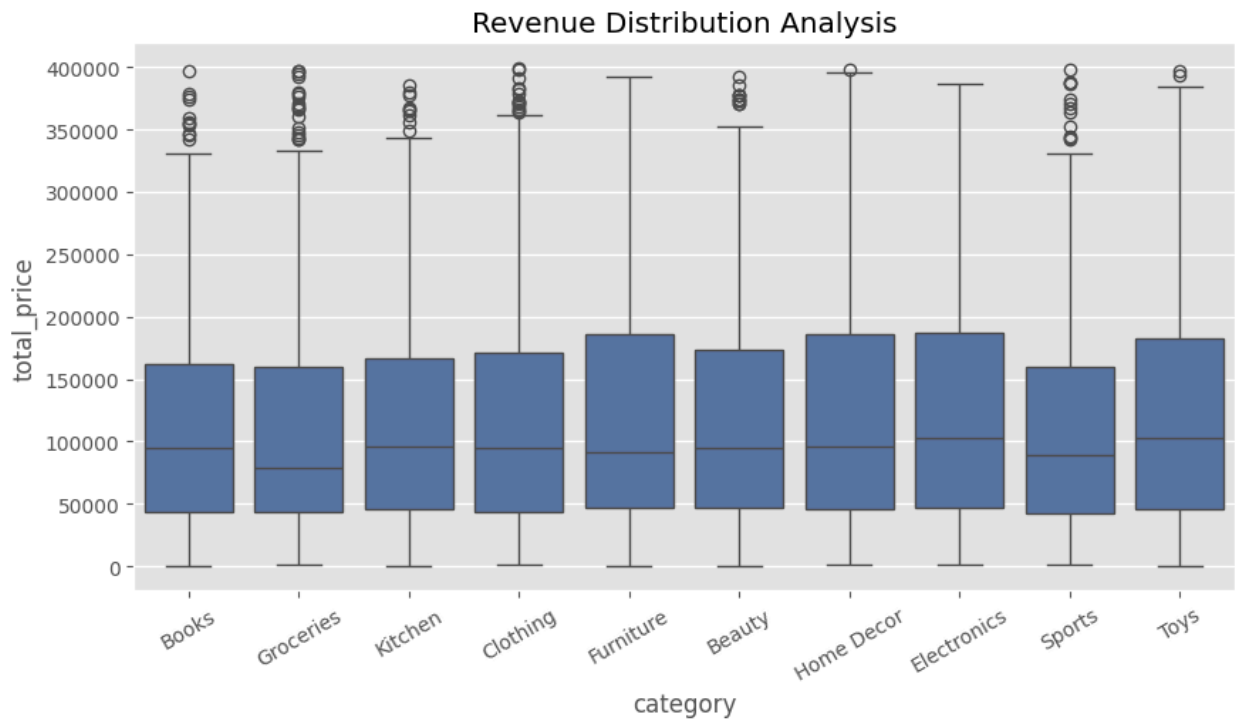
```
Out[21]: Text(0.5, 1.0, 'Quantity Purchased Analysis')
```



```
In [22]: #Insight  
#Most customers purchase products in small-to-medium quantities.  
#A small number of transactions contain significantly higher quantities, c
```

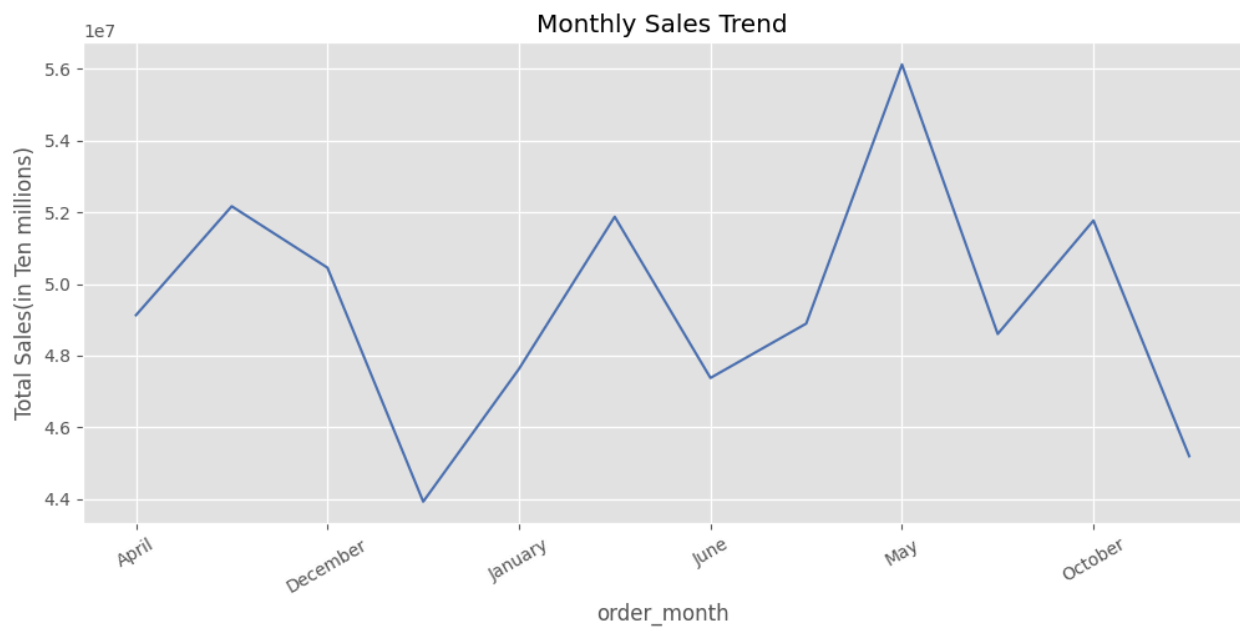
```
In [23]: # Multivariate Analysis
```

```
In [24]: plt.figure(figsize=(10,5))
sns.boxplot(x='category', y='total_price', data=df)
plt.title('Revenue Distribution Analysis')
plt.xticks(rotation=30)
plt.show()
```



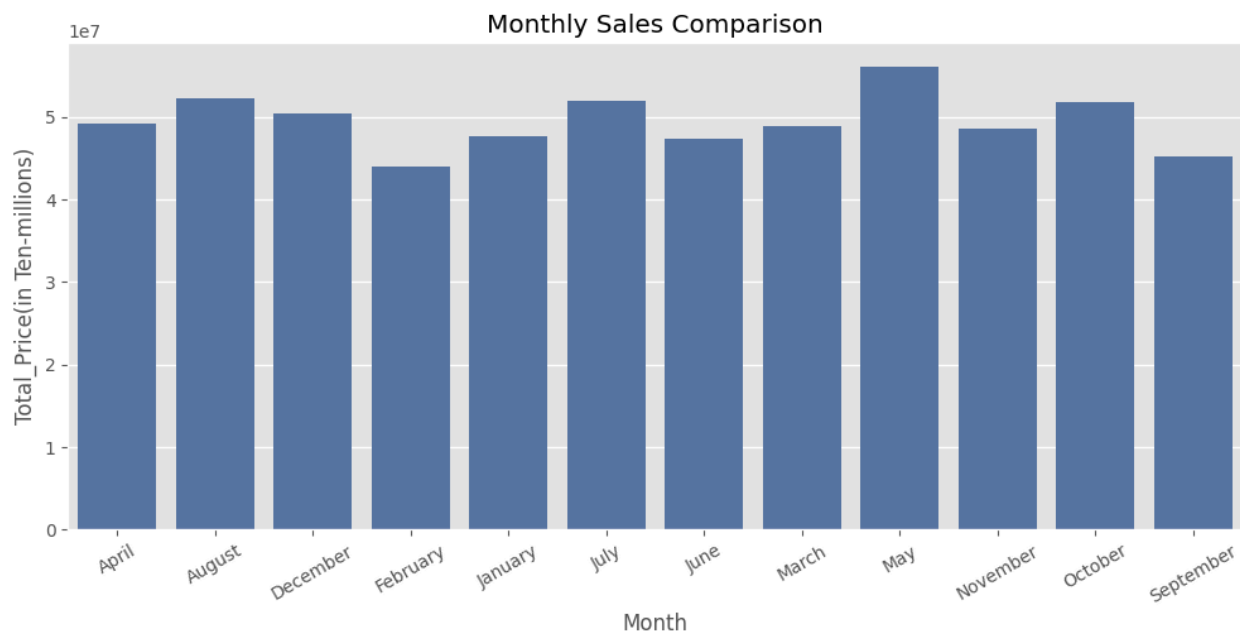
```
In [25]: #Insight
#The majority of customers generate average transaction values
#A smaller group of premium customers contributes a disproportionately high
```

```
In [26]: month_order = [
'January', 'February', 'March', 'April', 'May', 'June',
'July', 'August', 'September', 'October', 'November', 'December'
]
monthly_sales = df.groupby('order_month')['total_price'].sum()
monthly_sales.plot(figsize=(12,5))
plt.title('Monthly Sales Trend')
plt.ylabel('Total Sales(in Ten millions)')
plt.xticks(rotation=30)
plt.show()
```



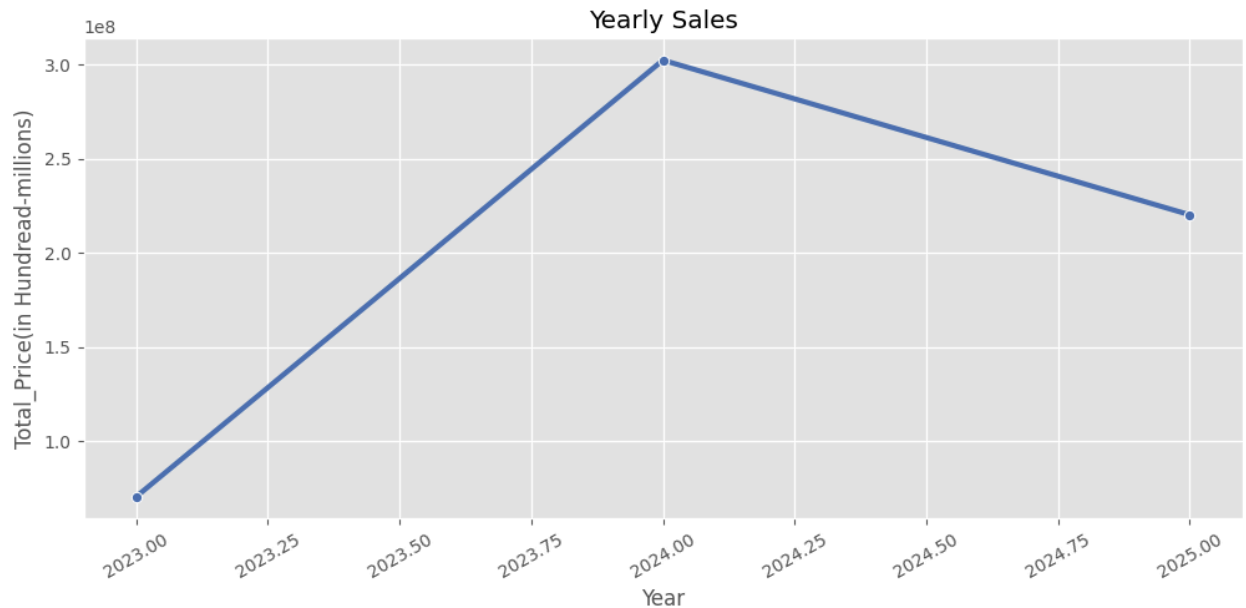
In [27]: *#Insights*
#May recorded the highest monthly sales revenue
#Certain months experienced lower revenue performance, indicating seasonal

In [28]: `monthly_sales = df.groupby('order_month')['total_price'].sum().reset_index()
plt.figure(figsize=(12,5))
sns.barplot(x='order_month',y='total_price',data=monthly_sales)
plt.title('Monthly Sales Comparison')
plt.xlabel('Month')
plt.ylabel('Total_Price(in Ten-millions)')
plt.xticks(rotation=30)
plt.show()`



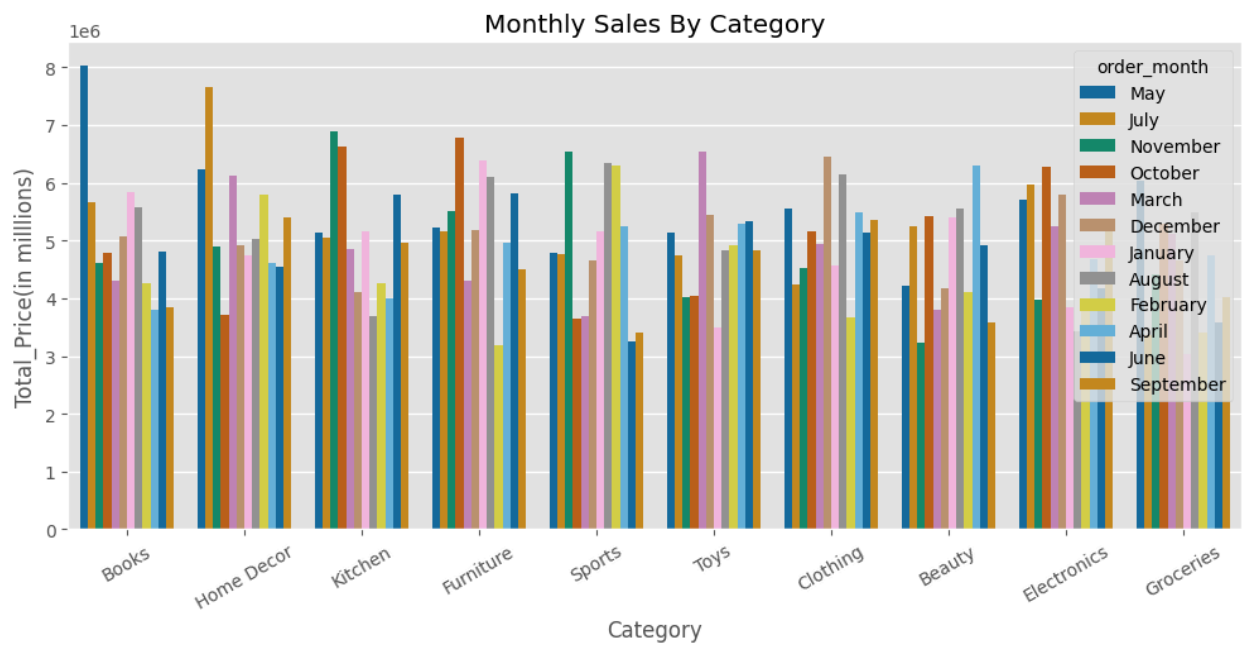
In [29]: `yearly_sales = df.groupby('order_year')['total_price'].sum().reset_index()`


```
plt.figure(figsize=(12,5))
sns.lineplot(x='order_year',y='total_price',data=yearly_sales,marker='o',
plt.title('Yearly Sales')
plt.xlabel('Year')
plt.ylabel('Total_Price(in Hundread-millions)')
plt.xticks(rotation=30)
plt.show()
```



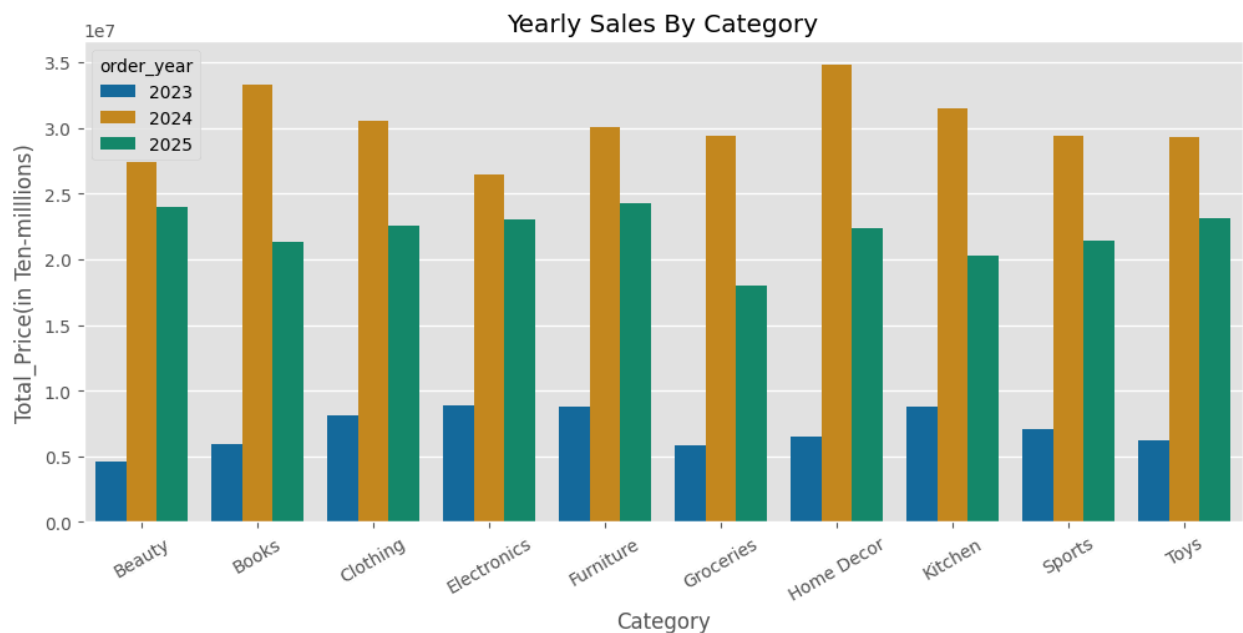
In [30]: *#Insight*
#The yearly sales trend shows strong business growth from 2023 to 2024, fo
while overall revenue remained significantly higher than the initial yea

In [31]: `monthly_sales = df.groupby(['category','order_month'])['total_price'].sum(
monthly_sales=monthly_sales.sort_values('total_price',ascending=False)
plt.figure(figsize=(12,5))
sns.barplot(x='category',y='total_price',data=monthly_sales,hue='order_mor
plt.title('Monthly Sales By Category')
plt.xlabel('Category')
plt.ylabel('Total_Price(in millllions)')
plt.xticks(rotation=30)
plt.show()`



In [32]: *#Insight*
Book category contributes higher in revenue, closely followed by Home De

```
In [33]: plt.figure(figsize=(12,5))
yearly_sales = df.groupby(['category','order_year'])['total_price'].sum().
sns.barplot(x='category',y='total_price',data=yearly_sales,hue='order_year')
plt.title('Yearly Sales By Category')
plt.xlabel('Category')
plt.ylabel('Total_Price(in Ten-milllions)')
plt.xticks(rotation=30)
plt.show()
```

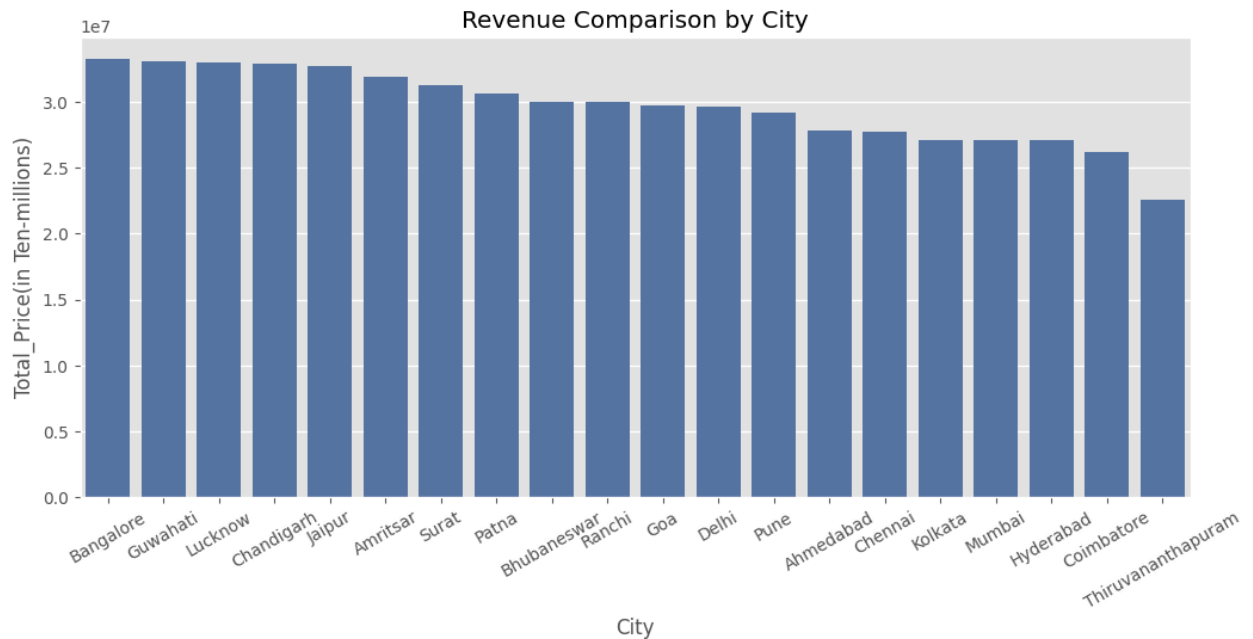


In [34]: *#Insight*

#Sales peaked across almost all categories in 2024—especially Home Decor and Groceries, with Groceries experiencing the sharpest drop.

In [35]:

```
monthly_sales = df.groupby('city')['total_price'].sum().reset_index()
monthly_sales=monthly_sales.sort_values('total_price',ascending=False)
plt.figure(figsize=(12,5))
sns.barplot(x='city',y='total_price',data=monthly_sales)
plt.title('Revenue Comparison by City')
plt.xlabel('City')
plt.ylabel('Total_Price(in Ten-millions)')
plt.xticks(rotation=30)
plt.show()
```

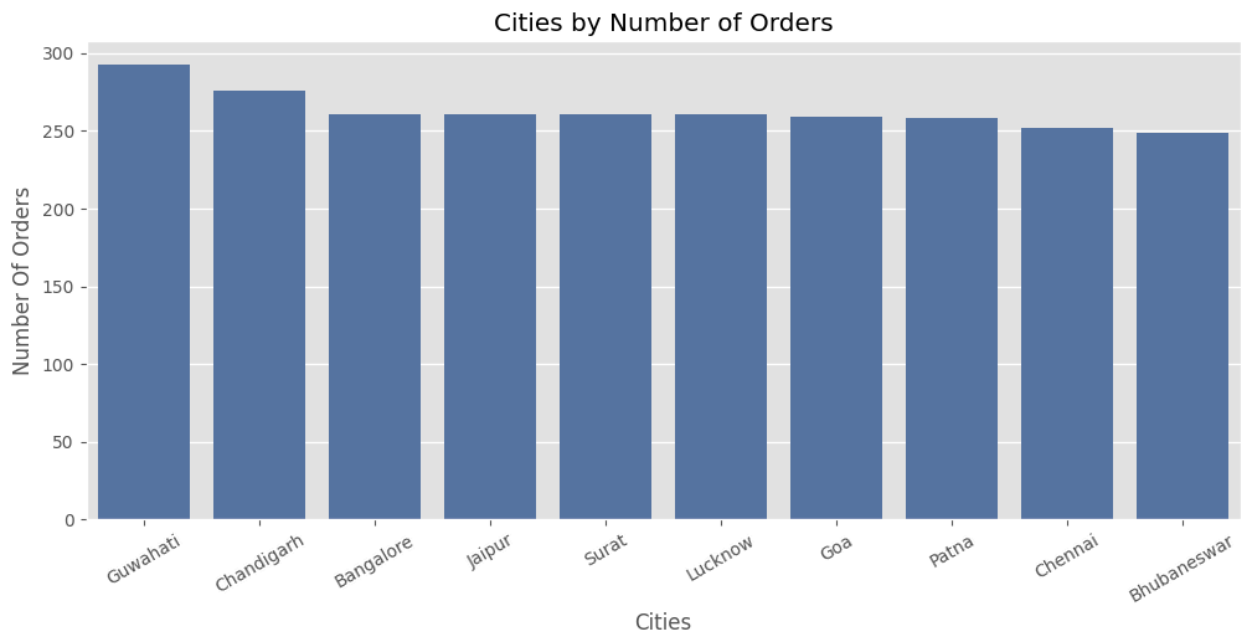


In [36]:

#Insight
#Bangalore generated the highest sales revenue among all cities, while Thiruvananthapuram generated the lowest.
#Overall, sales distribution across cities remained relatively balanced, indicating a widespread market presence.

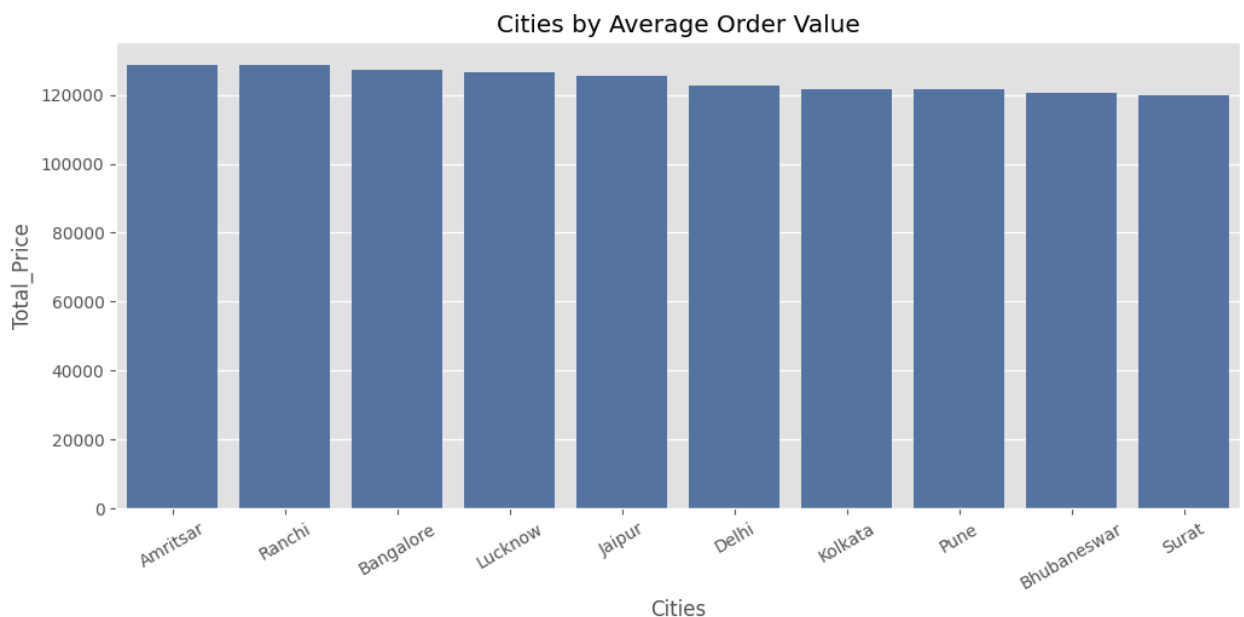
In [37]:

```
plt.figure(figsize=(12,5))
city_sales = df.groupby('city')['total_price'].count().reset_index()
city_sales = city_sales.sort_values('total_price', ascending=False).head(10)
sns.barplot(x='city',y='total_price',data=city_sales)
plt.title('Cities by Number of Orders')
plt.xlabel('Cities')
plt.ylabel('Number Of Orders')
plt.xticks(rotation=30)
plt.show()
```



In [38]: *#Insight*
#Guwahati shows higher number of orders, closely followed by Chandigarh and

```
In [39]: plt.figure(figsize=(12,5))
city_sales = df.groupby('city')['total_price'].mean().reset_index()
city_sales = city_sales.sort_values('total_price', ascending=False).head(10)
sns.barplot(x='city',y='total_price',data=city_sales)
plt.title('Cities by Average Order Value')
plt.xlabel('Cities')
plt.ylabel('Total_Price')
plt.xticks(rotation=30)
plt.show()
```



In [40]: *#Insight*

```
#AOV of Top cities are mainly same
```

```
In [41]: sns.heatmap(df.corr(numeric_only=True), annot=True)
```

```
Out[41]: <Axes: >
```



```
In [42]: #Insight  
#Total price is strongly positively correlated with price (0.72) and moder  
#while order year shows almost no relationship with the other variables.
```

Business KPIs

```
In [43]: #Total Revenue  
Total_Revenue=df['total_price'].sum()  
print(Total_Revenue)
```

```
593171089
```

```
In [44]: #Total Order  
df.shape[0]
```

```
Out[44]: 5000
```

```
In [45]: #Average Order Value  
AOV=df['total_price'].mean()  
print(AOV)
```

118634.2178

In [46]: *# Top City in Sales*

```
Top_City=df.groupby('city')['total_price'].sum().idxmax()  
print(Top_City)
```

Bangalore

In [47]: *# Top Month in Sales*

```
Top_Month=df.groupby('order_month')['total_price'].sum().idxmax()  
print(Top_Month)
```

May

In [48]: *#Top 5 Customers in Spending*

```
Top_Customers=df.groupby('customer_name')['total_price'].sum().sort_values(ascending=False)  
print(Top_Customers)
```

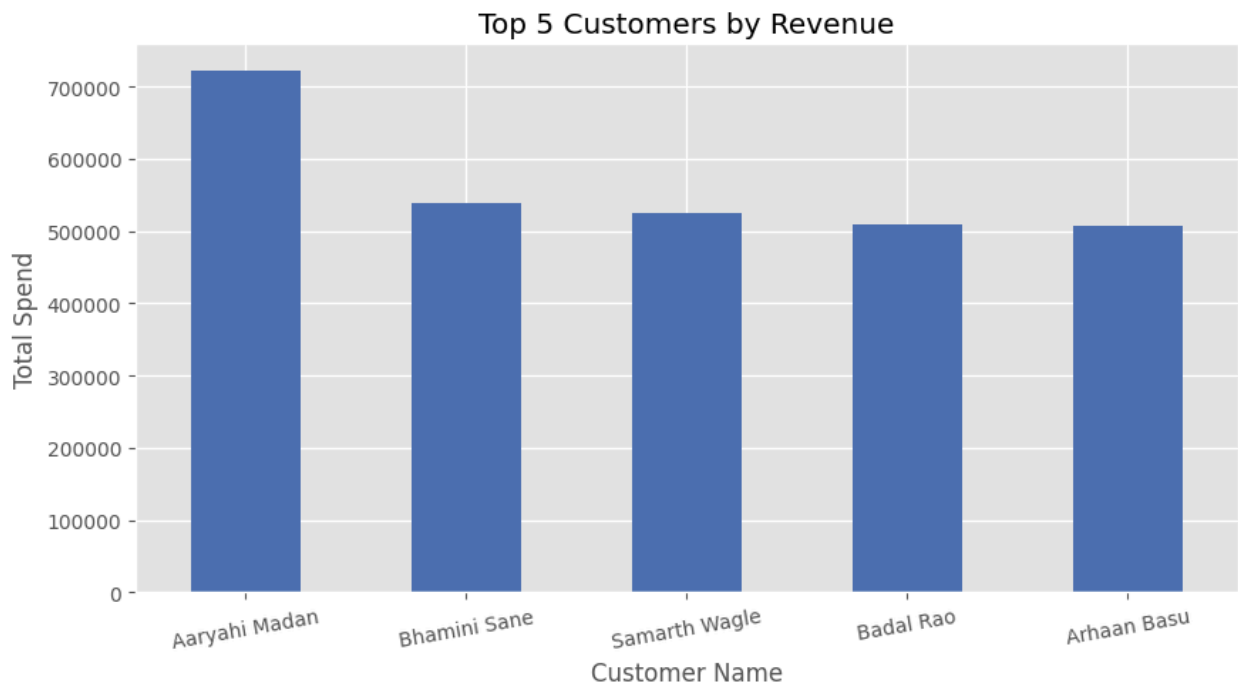
```
customer_name  
Aaryahi Madan    722391  
Bhamini Sane     538161  
Samarth Wagle    524785  
Badal Rao        509935  
Arhaan Basu      506272  
Name: total_price, dtype: int64
```

In [49]: `top_customers = df.groupby('customer_name')['total_price'].sum().sort_values(ascending=False)`

```
top_customers.plot(kind='bar', figsize=(10,5))
```

```
plt.title('Top 5 Customers by Revenue')  
plt.xlabel('Customer Name')  
plt.ylabel('Total Spend')  
plt.xticks(rotation=10)
```

```
plt.show()
```



Final Conclusion

--This E-Commerce Sales Analysis provides valuable insights into customer behavior, revenue performance, and business operations using transactional sales data. The analysis demonstrates that the company maintains: ->Strong revenue generation exceeding ₹593 million. ->Healthy customer purchasing power reflected through high AOV. ->Balanced category-wise revenue contribution. ->The findings also indicate that a relatively small percentage of premium customers contributes a substantial portion of total revenue, highlighting opportunities for customer retention and loyalty-based strategies.

Recommended Business Actions

Based on the analysis, the company should focus on:- ->Expanding operations in high-performing cities ->Improving inventory management for top-selling products ->Launching loyalty programs for high-value customers ->Implementing targeted customer segmentation strategies ->Using seasonal demand insights for better planning

In []: